

# DATA SCIENCE TRAINING

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# PROJECT-1

## Sales Data Analysis Excel Dashboard

\*"This dashboard contains a pivot table and multiple charts. The pivot table summaries yearly sales, while the charts visually compare the performance of each sales representative across different years."

\*"The total sales across all three years are **\$19.2 million**. This represents the combined sales generated by all three sales representatives."

"Looking at yearly performance:

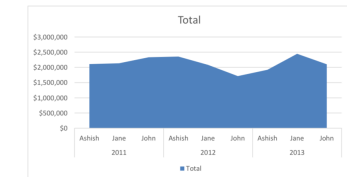
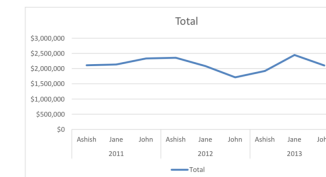
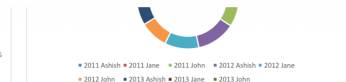
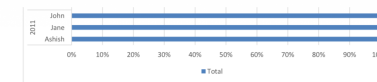
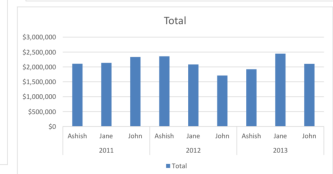
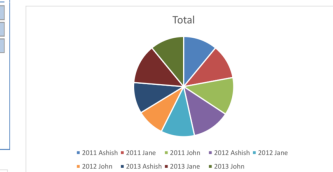
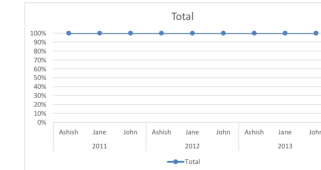
- In **2011**, total sales were **\$6.58 million**.
- In **2012**, total sales slightly decreased to **\$6.15 million**.
- In **2013**, sales improved again to **\$6.47 million**."

"The bar charts make it easy to compare sales values between employees."

"The percentage chart shows each representative's contribution to total yearly sales."

"These visualisation help identify trends and compare performance without reading large tables."

| Year        | Sales_Rep_Name | Sum of Value |
|-------------|----------------|--------------|
| 2011        | Ashish         | \$2,107,310  |
|             | Jane           | \$2,135,825  |
|             | John           | \$2,333,251  |
| 2011 Total  |                | \$6,576,386  |
| 2012        | Ashish         | \$2,256,655  |
|             | Jane           | \$2,082,686  |
|             | John           | \$1,712,599  |
| 2012 Total  |                | \$6,151,940  |
| 2013        | Ashish         | \$1,922,074  |
|             | Jane           | \$2,446,759  |
|             | John           | \$2,102,303  |
| 2013 Total  |                | \$6,471,136  |
| Grand Total |                | \$19,199,461 |



# Project-2

"At the top, we have the Dashboard Overview section, which displays important Key Performance Indicators (KPIs)."

On the left side, we have interactive filters that allow users to customise the analysis.

- Gender Filter
- Age Range Slider
- Smoking Status
- Clinical Diagnosis
- Cardiac Risk Level

## Graphs

- \*Gender Distribution
- \*Cardiac Risk Level Distribution
- \*BMI Distribution
- \*Blood Pressure Distribution
- \*Diagnosis vs Risk Level

## Patient Records Table

At the bottom, we have the Patient Records table, which displays detailed information for the first 20 patients." The table includes:

- Patient ID
- Age
- Gender
- BMI
- Blood Pressure
- Smoking Status
- Diagnosis
- Risk Level



| Top 20 Patient Records |     |        |      |             |              |         |                               |            |
|------------------------|-----|--------|------|-------------|--------------|---------|-------------------------------|------------|
| Patient ID             | Age | Gender | BMI  | Systolic BP | Diastolic BP | Smoking | Diagnosis                     | Risk Level |
| P000001                | 68  | Male   | 48.0 | 131         | 86           | Former  | Mild Arrhythmia / At Risk     | Low        |
| P000002                | 81  | Female | 32.7 | 187         | 125          | Never   | Hypertensive Heart Disease    | Medium     |
| P000003                | 58  | Female | 29.7 | 166         | 105          | Never   | Mild Arrhythmia / At Risk     | Low        |
| P000004                | 44  | Male   | 40.0 | 118         | 67           | Current | Healthy / No Significant Risk | Low        |
| P000005                | 72  | Female | 25.8 | 124         | 81           | Never   | Hypertensive Heart Disease    | Medium     |
| P000006                | 37  | Male   | 36.1 | 134         | 92           | Former  | Hypertensive Heart Disease    | Medium     |
| P000007                | 90  | Male   | 45.8 | 163         | 101          | Never   | Mild Arrhythmia / At Risk     | Low        |
| P000008                | 50  | Female | 18.9 | 183         | 108          | Current | Hypertensive Heart Disease    | Medium     |
| P000009                | 68  | Male   | 22.4 | 179         | 126          | Current | Coronary Artery Disease (CAD) | High       |
| P000010                | 87  | Female | 28.1 | 188         | 118          | Never   | Hypertensive Heart Disease    | Medium     |
| P000011                | 48  | Male   | 38.5 | 181         | 109          | Never   | Hypertensive Heart Disease    | Medium     |
| P000012                | 52  | Female | 23.9 | 118         | 85           | Former  | Mild Arrhythmia / At Risk     | Low        |
| P000013                | 40  | Female | 15.9 | 159         | 97           | Never   | Coronary Artery Disease (CAD) | High       |
| P000014                | 40  | Male   | 16.6 | 92          | 51           | Former  | Healthy / No Significant Risk | Low        |
| P000015                | 53  | Female | 19.3 | 182         | 122          | Current | Mild Arrhythmia / At Risk     | Low        |
| P000016                | 82  | Male   | 14.6 | 146         | 89           | Former  | Hypertensive Heart Disease    | Medium     |
| P000017                | 65  | Male   | 34.3 | 155         | 104          | Current | Hypertensive Heart Disease    | Medium     |
| P000018                | 69  | Male   | 25.6 | 111         | 64           | Never   | Healthy / No Significant Risk | Low        |
| P000019                | 53  | Female | 29.6 | 116         | 88           | Current | Mild Arrhythmia / At Risk     | Low        |

# Project 3

## AI Industry Investment Analysis Dashboard

### Tools Used

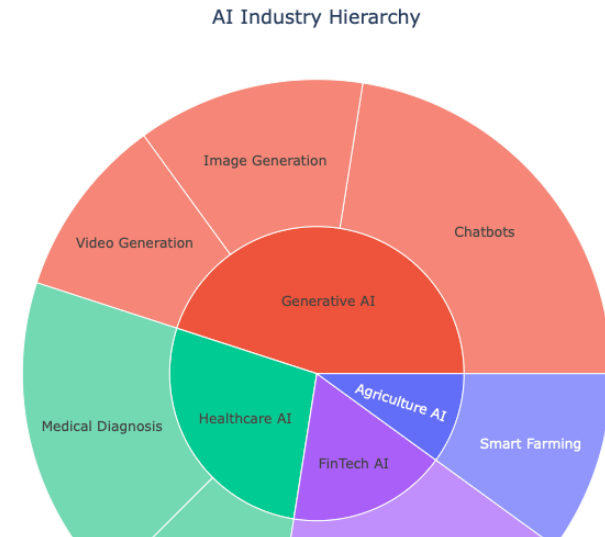
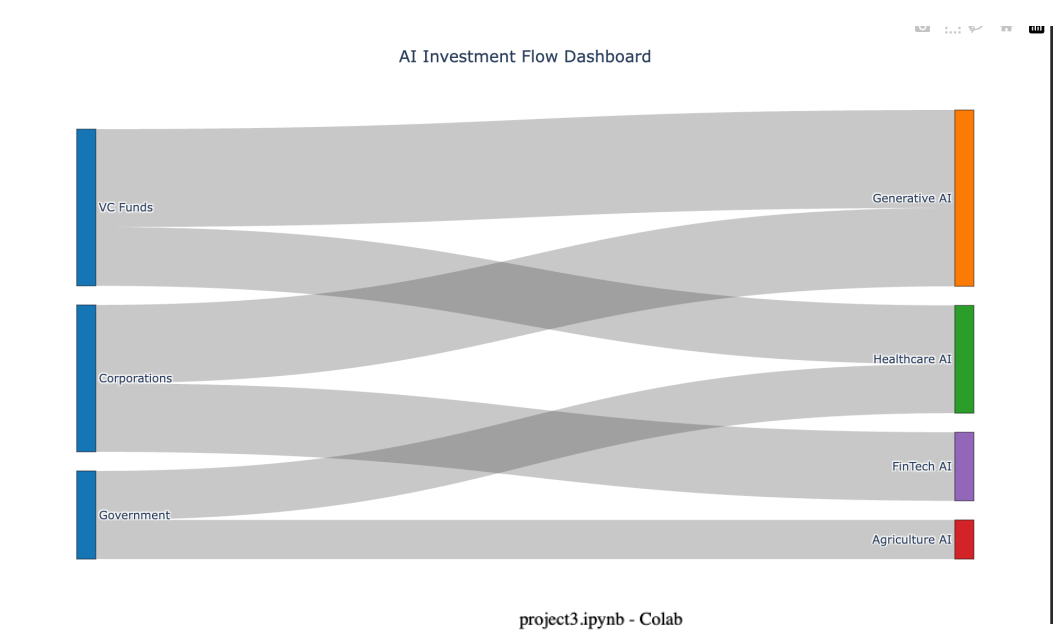
- Python
- Google Colab
- Pandas
- Plotly Graph Objects
- Plotly Express

### Sankey Diagram

"The Sankey Diagram represents the flow of investments from different funding sources to AI sectors."

### Sunburst Chart

"The Sunburst Chart displays the hierarchical relationship between AI sectors and their applications."



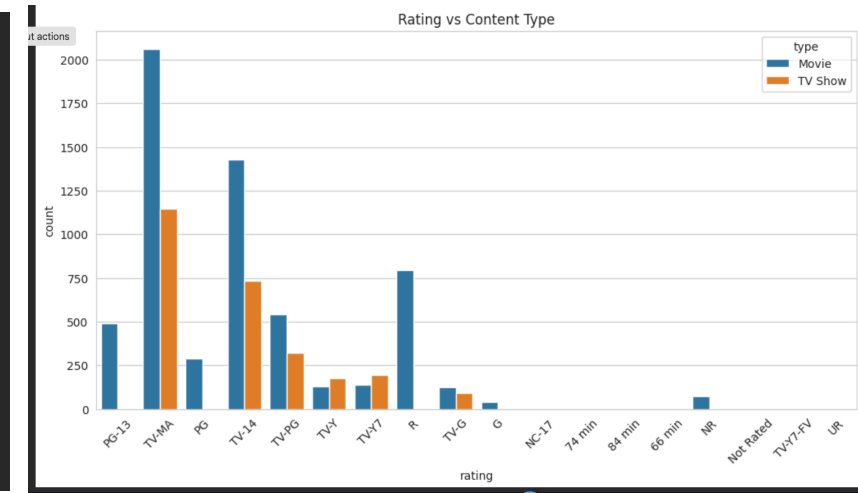
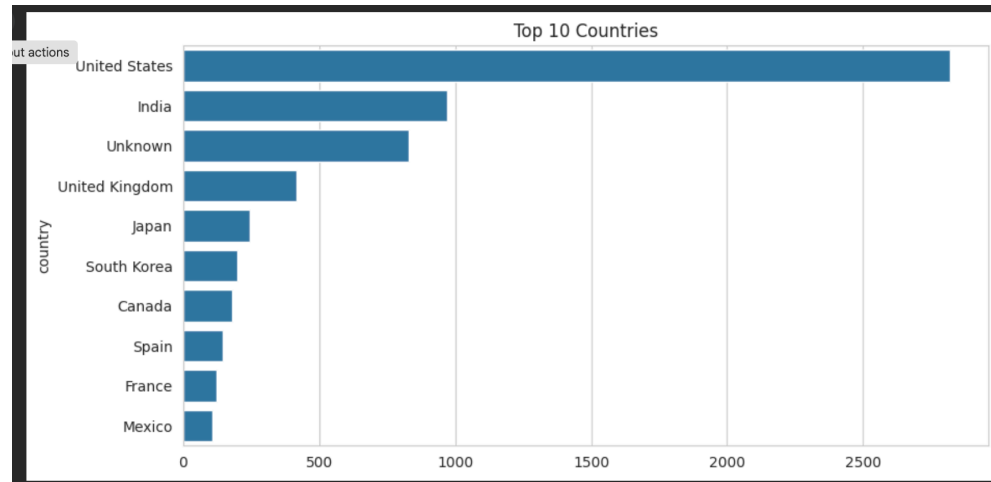
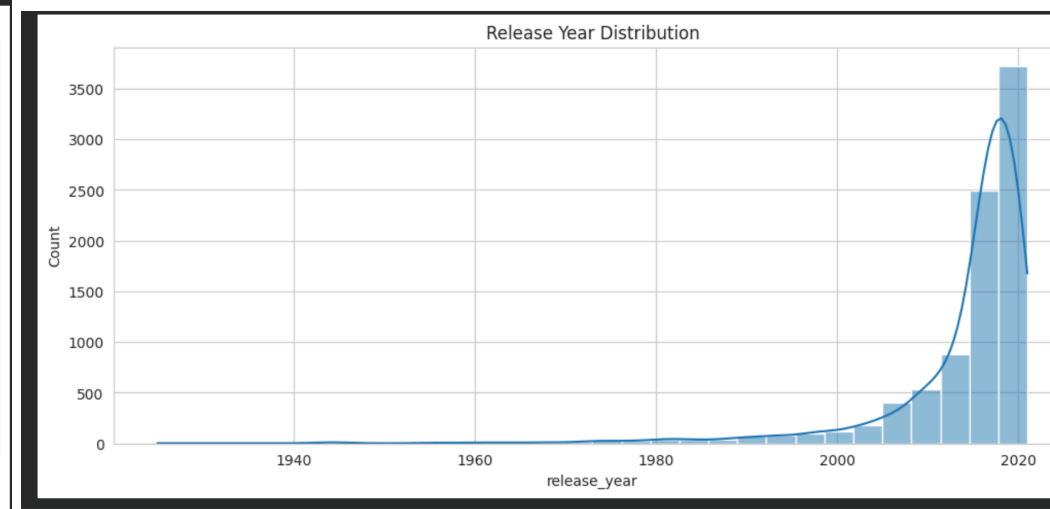
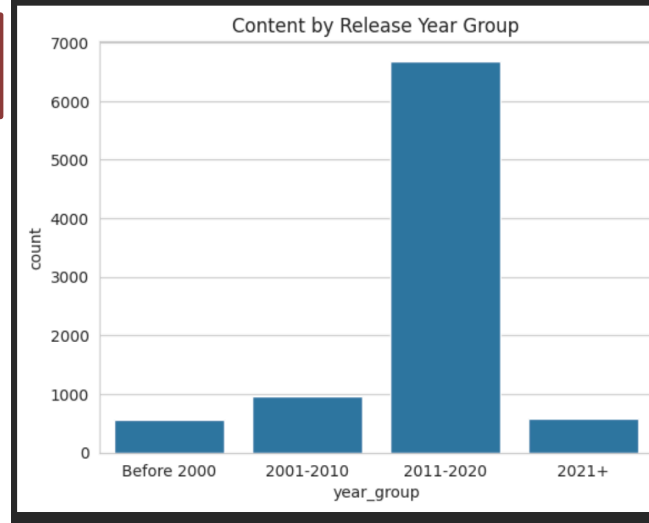
## Netflix Content Analytics

“This dashboard provides an interactive summary of Netflix content. It includes charts and key performance indicators (KPIs) that help us understand content distribution by type, release year, rating, country, and genre.”

“This chart compares the number of Movies and TV Shows available on Netflix. It helps us understand which type of content dominates the platform.”

“This chart displays which countries have contributed the most titles to Netflix. It helps identify Netflix's major content-producing regions.”

“This chart shows the distribution of content ratings such as TV-MA, TV-14, PG-13, and others. It helps us understand the target audience.”



Github- <https://github.com/Himanshu3557/himanshu3557/blob/main/Netflix.ipynb>

# Project-5

## “Overview”

E-commerce Sales Dashboard developed using Microsoft Power BI. The main objective of this dashboard is to analyse sales performance, identify business trends, compare payment methods, and support data-driven decision-making through interactive visualisation.

"At the top of the dashboard, I have displayed four important Key Performance Indicators (KPIs)."

- \*Total Orders
- \*Maximum Sale
- \*Average Sales
- \*Minimum Sale

### #Stacked Column Chart

"This chart compares sales across different product categories while also showing which payment methods customers used."

### #Sales by Payment Mode (Donut Chart)

"This donut chart shows the percentage contribution of each payment method."

### #100% Stacked Column Chart

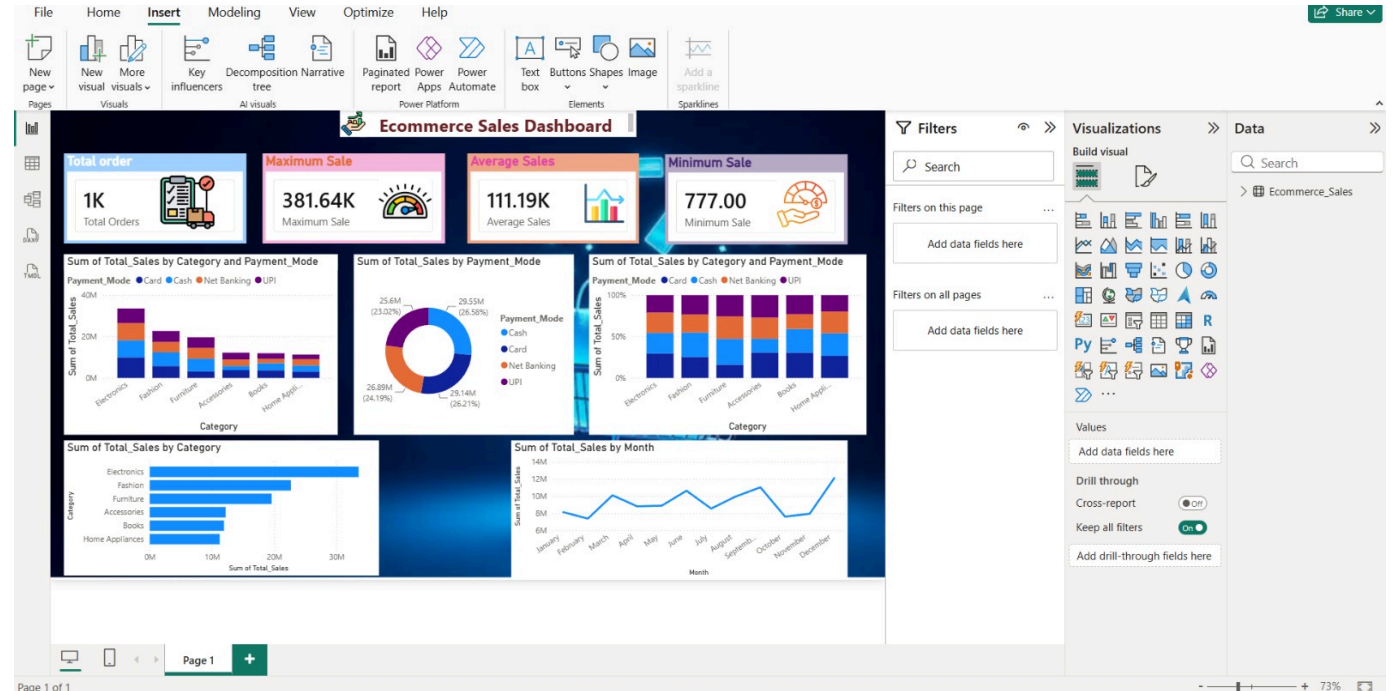
"This visualisation shows the percentage distribution of payment methods for each product category."

### #Sales by Category (Horizontal Bar Chart)

"This chart ranks product categories based on total sales."

### #Monthly Sales Trend (Line Chart)

"This line chart displays sales performance month by month."



## Tools Used

- **Microsoft Power BI Desktop** – Used to create interactive dashboards and visualizations.
- **Microsoft Excel** – Used as the data source for sales data.
- **Power Query Editor** – Used for data cleaning, removing duplicates, handling missing values, and transforming data.
- **DAX (Data Analysis Expressions)** – Used to create calculated measures, KPIs, and perform data analysis.
- **Data Modeling** – Used to establish relationships between tables (if multiple tables were used).

# Thank You

